Supplementary Materials

S1: Replication of Experiment 1

We sought to replicate Experiment 1 using an in-lab task. As in Experiment 1, the central question was whether adults would make selections that reduced referential ambiguity. In addition to replicating the result from Experiment 1, we explored the degree to which adults' sampling decisions were sensitive in a graded fashion to the extent of ambiguity in their options. We manipulated the degree of ambiguity across two between-subjects conditions. One condition (the Fully Ambiguous condition) matched the design of Experiment 1, such that one set of items remained fully ambiguous throughout the training phase, while object-label associations were disambiguated for the remaining items. In the second condition (the Partially Ambiguous condition), one set of items was manipulated to remain moderately ambiguous across training, while the remaining items were disambiguated as in the Fully Ambiguous condition.

As in Experiment 1, we predicted that participants would be more likely to choose to learn about the ambiguous items than about the disambiguated items in the sampling phase. For the Partially Ambiguous condition, we expected participants to have a weaker preference for ambiguous items over the disambiguated items, since adults accurately tracking the co-occurrence evidence could successfully learn all word-referent pairs.

Method

Participants

62 University undergraduates at a large public university in the Midwestern United States (27 female; mean age: 19.1 years, SD = 1.01; 56 native speakers of English) participated for course credit and were randomly assigned to the Fully Ambiguous Condition (n = 28) or the Partially Ambiguous Condition (n = 34).

Stimuli

The stimuli were identical to Experiment 1. All stimuli and the experimental scripts are openly available on the Open Science Framework [https://osf.io/udmvh/].

Design & Procedure

The experiment consisted of a Training Phase, a Sampling Phase, and a Test Phase. In Experiment S1 (and unlike Experiment 1), participants completed the Test Phase twice, once after the Training Phase and once after the Sampling Phase (i.e., the order was Training-Test-Sampling-Test).

Training Phase. As in Experiment 1, participants viewed 24 cross-situational learning trials (2 blocks of 12 trials), presented in random order. The design and procedure were identical to Experiment 1, with the following exceptions.

Participants were assigned to one of two conditions: Fully Ambiguous or Partially Ambiguous. The Fully Ambiguous condition was identical to the design of Experiment 1, where half of the object-label pairs were never disambiguated (ambiguous items; Figure S1.1, top left) and half of the object-label pairs were disambiguated across training trials (disambiguated items; Figure S1.1, right panel). In the Partially Ambiguous condition, half of the object-label pairs were grouped such that two specific objects co-occurred on 4 out of their 6 occurrences, but each object occurred with a different object from the ambiguous object set on the remaining 2 trials (partially ambiguous items; Figure S1.1, bottom left). The other four objects were disambiguated as in the Fully Ambiguous condition. In both conditions, participants saw each individual object and label equally frequently.

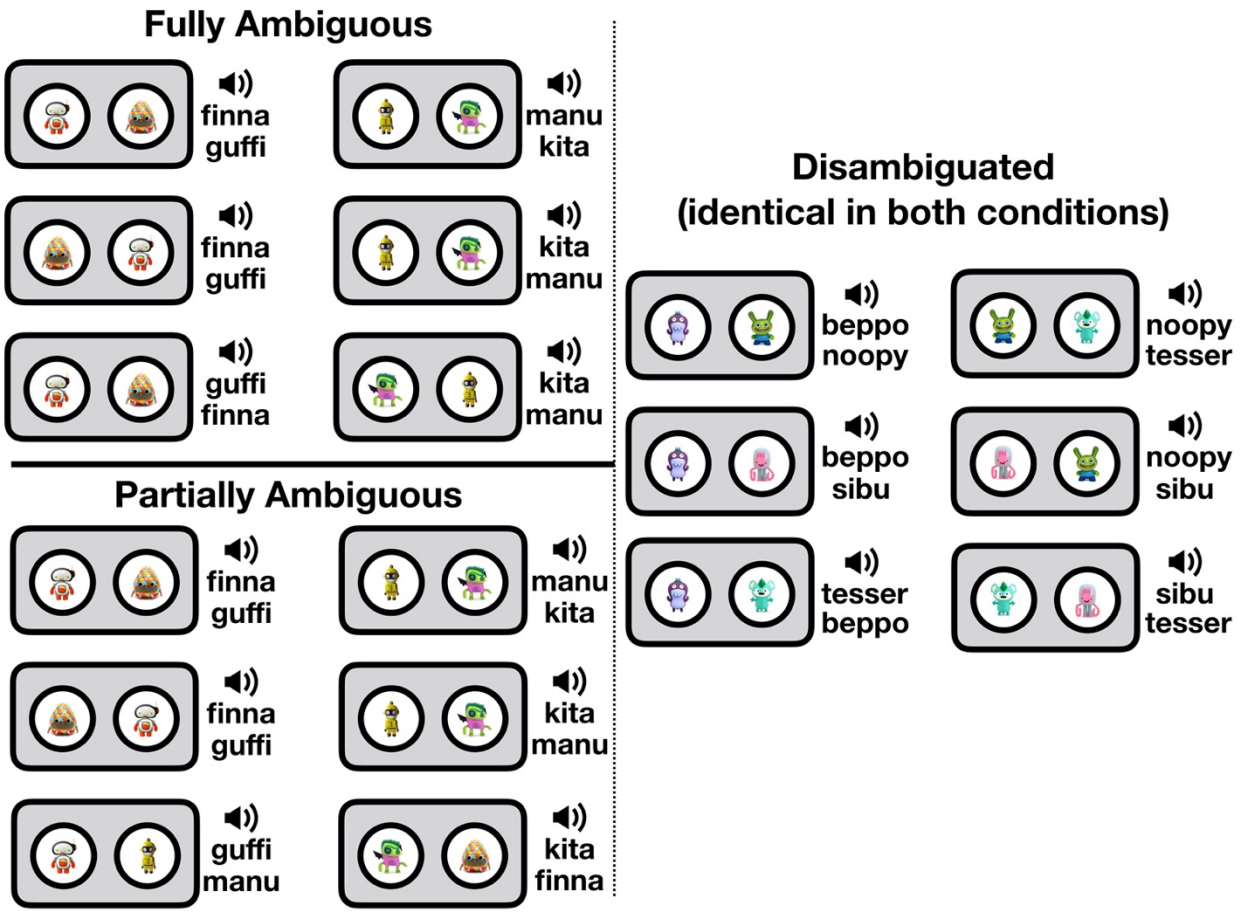


Figure S1.1. Overview over one block of the Training Phase for the Fully Ambiguous Condition and the Partially Ambiguous Condition

Sampling Phase. The procedure for the Sampling Phase was identical to the Sampling Phase in Experiment 1 (and identical in both conditions).

Test Phase. Knowledge of the object-label associations was tested in the same manner as in Experiment 1, with the exception that the 8-AFC test phase was presented twice: once immediately following the Training Phase (Test Block 1) and once immediately following the Sampling Phase (Test Block 2), as in Experiment 1. Our rationale in testing participants twice was to further assess the consequences of participants' sampling behavior for word learning. By testing object-label knowledge before and after the Sampling Phase, we hoped to investigate whether participants' choices led to gains in test accuracy. While adding a test phase immediately following

the Training Phase provided participants with additional exposure to the labels and the objects prior to the Sampling Phase, we reasoned that the additional test phase (Test Block 1) should not help participants learn the particular object-label associations, since all eight objects were presented with each label (i.e., no cross-situational learning could occur during the test phase) and participants did not receive feedback on their test choices.

Results

Sampling choices

We fit the same logistic mixed-effects model as in Experiment 1, with the addition of a fixed effect for condition (centered; fully ambiguous = 0.5, partially ambiguous = -0.5). To test whether participants' likelihood of selecting an ambiguous item differed from chance within each condition, we refit the model while recoding the condition variable with each condition coded as zero. Participants were significantly more likely to select a (fully or partially) ambiguous item in the Fully Ambiguous condition (M = 64.3%, 95% CI = [53.6%, 75.0%]) than in the Partially Ambiguous condition (M = 47.8%, 95% CI = [39.1%, 56.5%]), $b = 0.70$, 95% Wald CI = [0.14, 1.25], $z = 2.47$, $p = .01$ (Figure S1.2). Participants in the Fully Ambiguous condition selected ambiguous items more frequently than disambiguated items ($b = 0.60$, 95% CI = [0.19, 1.02], $z = 2.85$, $p = .004$), replicating the sampling preference from Experiment 1. Participants in the Partially Ambiguous condition did not select (partially) ambiguous items more frequently than disambiguated items ($b = -0.09$, 95% CI = [-0.45, 0.27], $z = -0.50$, $p = .62$).

We conducted non-parametric statistical analyses to test the robustness of the results in the Sampling Phase. A Wilcoxon rank-sum test comparing subjects' proportion of ambiguous selections between the Fully Ambiguous condition and the Partially Ambiguous condition was significant, $W = 632$, $p = .02$. Participants' average proportion of ambiguous selections

significantly diverged from chance in the Fully Ambiguous condition ($V = 140$, $p = .015$), but not in the Partially Ambiguous condition ($V = 101$, $p = .59$).

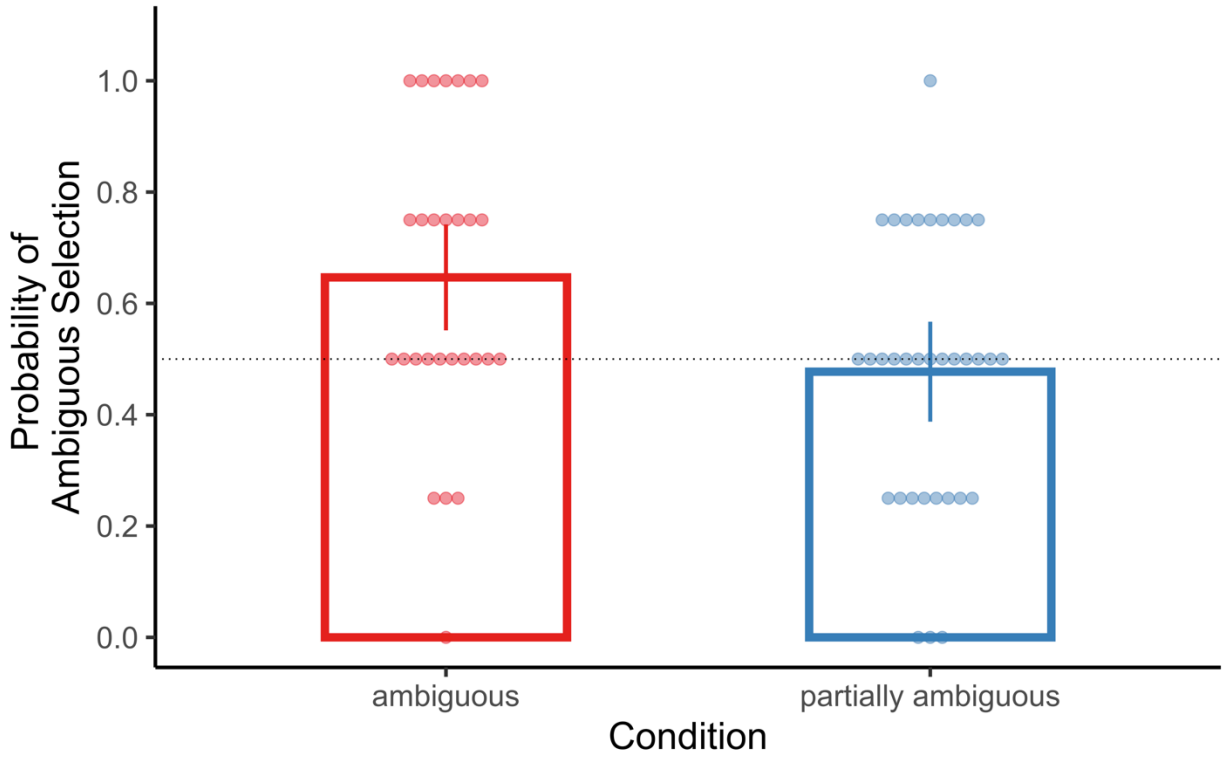


Figure S1.2. Proportion of more ambiguous items selected in each condition. Error bars in represent the 95% CI of the model predictions.

Test performance

Overall test accuracy. Participants demonstrated successful word learning in the 8-AFC recognition test that immediately followed the Training Phase (Fully Ambiguous condition: M = 65.2%, 95% CI = [61.2%, 69.2%]; Partially Ambiguous condition: M = 76.8%, 95% CI = [72.5%, 81.2%]) and in the (identical) test phase following the Sampling Phase (Fully Ambiguous condition: M = 72.8%, 95% CI = [68.8%, 76.8%]; Partially Ambiguous condition: M = 77.6%, 95% CI = [73.2%, 81.9%]).

Effects of Test Half and Item Type. To investigate the relationship between Test Half, Item Type (ambiguous vs. disambiguated), and Condition, we fit a logistic mixed-effects model

predicting trial-by-trial accuracy from the three-way interaction between these three predictors (centered), including all lower-order effects. We included by-participant and by-item random intercepts, and by-participant random slopes for Test Block, Item Type, and their interaction. There was a significant effect of Item Type (b = -1.33, Wald 95% CI = [-2.10, -0.56], z = -3.37, p < .001), indicating that participants generally performed better on items disambiguated during training across conditions, and a significant effect of Test Block (b = 0.82, Wald 95% CI = [0.11, 1.53], z = 2.27, p = .02), indicating that participants performed better on the second test block than the first. This effect of Test Block appeared to be driven mainly by a significant increase in accuracy for ambiguous items in the Fully Ambiguous condition from Test Block 1 M = 46.4%, 95% CI = [37.8%, 55.1%]) to Test Block 2 (M = 59.8%, 95% CI = [51.6%, 68.1%]), b = 1.04, Wald 95% CI = [0.18, 1.89], z = 2.38, p = .017 (see Figure S1.3A). There was also a significant interaction between Item Type and Condition, indicating that the difference in accuracy between disambiguated and ambiguous items was greater in the Fully Ambiguous condition than in the Partially Ambiguous condition (averaging across test block), b = -2.75, Wald 95% CI = [-3.87, -1.64], z = -4.83, p < .001 (Figure S1.3). All other effects, including the three-way interaction, were non-significant, all ps > .25.

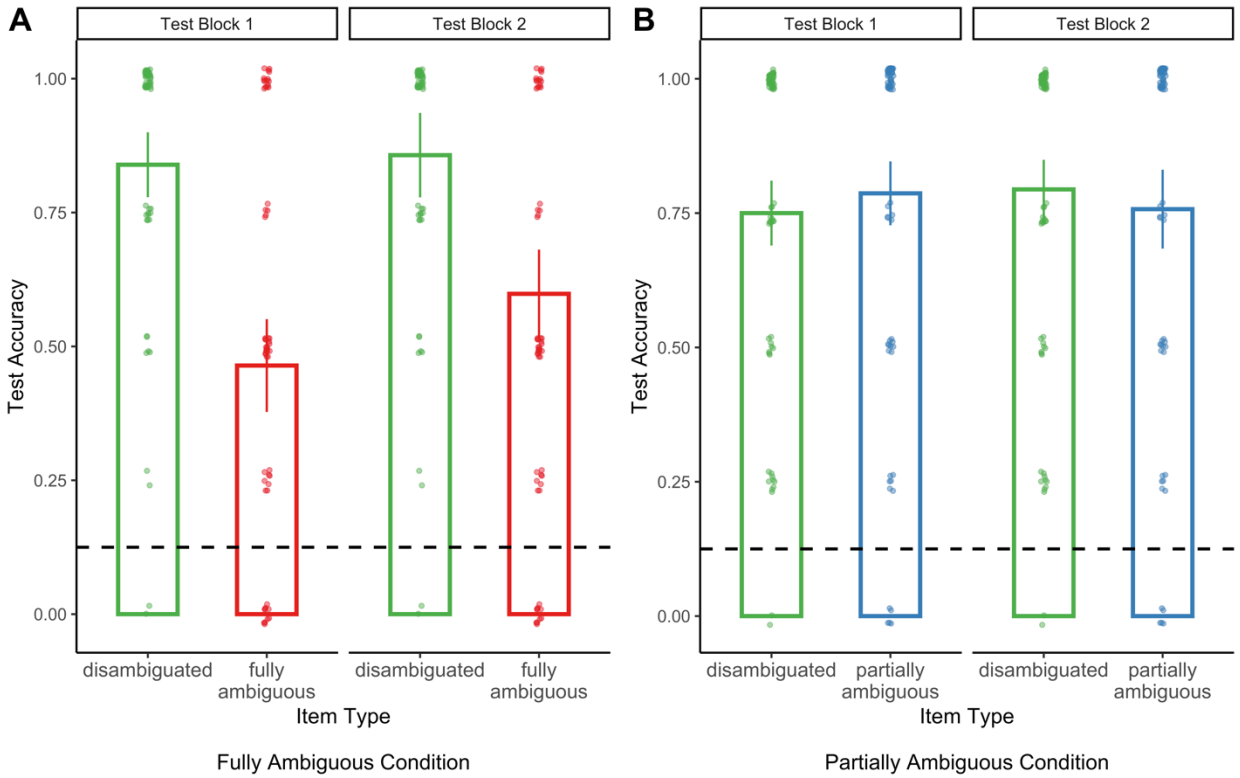


Figure S1.3. Test accuracy by item for (A) the Fully Ambiguous condition and (B) the Partially Ambiguous condition in Experiment S1. The dashed line represents chance-level performance. Error bars represent within-participant 95% CIs (Morey, 2008).

Relationship between sampling and test accuracy. In the Fully Ambiguous condition, participants' proportion of ambiguous selections was correlated with their test accuracy before the Sampling Phase (Test Block 1), $r(26) = .45$, 95% CI = [.10, .71], $p = .015$. The proportion of ambiguous selections was marginally correlated with participants' test accuracy after the Sampling Phase (Test Block 2), $r(26) = .35$, 95% CI = [-.03, .64], $p = .07$ (Figure S1.4, panel A). In the Partially Ambiguous condition, participants' proportion of (partially) ambiguous items selected was not significantly correlated with their test accuracy before (Test Block 1: $r(32) = .10$, 95% CI = [-.25, .42], $p = .58$) or after the Sampling Phase (Test Block 2: $r(32) = -.12$, 95% CI = [-.44, .23], $p = .50$; Figure S1.4, panel B).

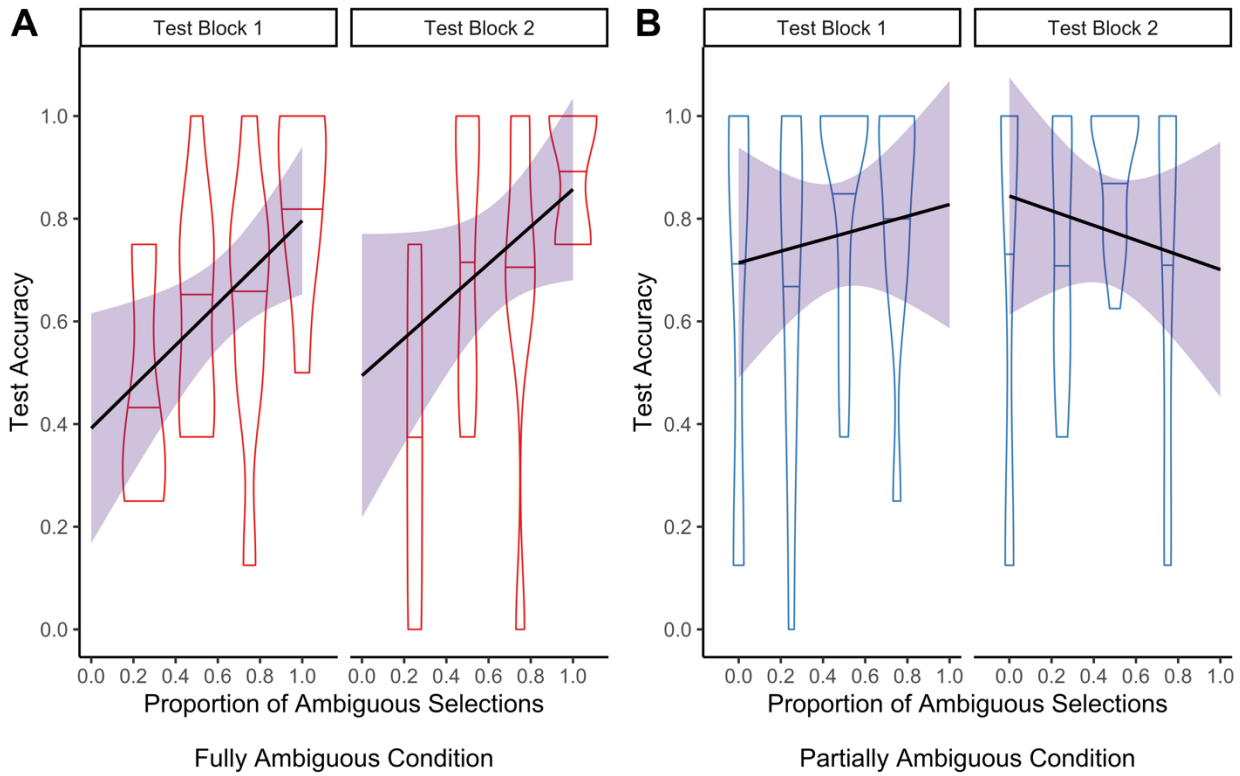


Figure S1.4. Relationship between choosing more ambiguous items and test accuracy for (A) the Fully Ambiguous condition and (B) the Partially Ambiguous condition in Experiment S1, split by Test Block. The distribution of individual participants' test accuracy for different proportions of ambiguous choices is represented with violin plots. Error bands represent 95% CIs.

By including a test phase immediately preceding the Sampling Phase, we aimed to further understand the correlation between test accuracy and preference for selecting ambiguous items observed in Experiment 1. Specifically, do participants have higher test accuracy at the conclusion of the experiment because they preferentially selected ambiguous items, or do participants who are more successful at learning the object-label associations show a stronger preference for selecting ambiguous items?

To address this question, we correlated participants' proportion of ambiguous selections with their increase in accuracy from Test Block 1 to Test Block 2. If participants' selections of ambiguous items are driving higher accuracy, then participants who show a preference for sampling ambiguous items should show the largest increases in accuracy from Test Block 1 to

Test Block 2. However, proportion of ambiguous items selected was not significantly correlated with an increase in test accuracy in the Fully Ambiguous condition ($r(26) = -.08$, 95% CI = [-.44, .30], $p = .69$) and was negatively correlated in the Partially Ambiguous condition ($r(32) = -.36$, 95% CI = [-.62, -.03], $p = .04$), i.e. participants who were more likely to select the (partially) ambiguous items showed a lesser increase in test accuracy. Similar relationships were found with test accuracy for both ambiguous and disambiguated items.

Discussion

In a cross-situational learning task, adult learners chose to learn more about those object-label pairs that remained ambiguous throughout training. Adults showed this tendency when the object-label pairings were truly ambiguous based on the training evidence (Fully Ambiguous condition), but not when the object-label pairs became disambiguated at any point during training (Partially Ambiguous condition). This experiment replicates the result from Experiment 1 (in the Fully Ambiguous condition) and provides further evidence that adult learners will seek to reduce ambiguity about object-label associations, primarily when word referents remain entirely ambiguous from past experience.

At test, participants tended to show poorer overall learning of the ambiguous object-label pairs in the Fully Ambiguous condition, though not in the Partially Ambiguous condition. This finding is consistent with the fact that the fully ambiguous items were (by definition) more difficult object-label mappings to learn (though note that we only found a marginal difference in accuracy between item types in Experiment 1). Intriguingly, we also found that participants' sampling behavior was correlated with their test accuracy in the Fully Ambiguous condition: participants who chose more ambiguous items during the Sampling Phase also more accurately identified object-label associations. By testing participants before and after the Sampling Phase

in Experiment S1, we were able to partially disentangle the directionality of this effect. Participants who had learned the novel words better following the Training Phase were more likely to sample ambiguous items. In other words, participants' learning success appeared to predict their likelihood of targeting ambiguous items during the Sampling Phase, rather than participants' sampling choices (solely) driving their test accuracy. Perhaps surprisingly, participants' likelihood of selecting ambiguous items did not appear to lead to larger increases in accuracy from the first to the second testing (though accuracy for ambiguous items showed the largest increase in general).

There are a few potential reasons why we did not observe greater increases in accuracy following ambiguous item selections in the present study. First, participants who showed a strong preference for selecting ambiguous items were already performing quite well on the test, suggesting that the lack of an effect may in part be due to a ceiling effect. Second, the Sampling Phase was designed such that participants continued to receive ambiguous, cross-situational learning trials: even after selecting ambiguous items, participants would view a cross-situational learning trial involving an additional, randomly selected item. While these trials in principle provided the opportunity to disambiguate object-label mappings, they also required participants to simultaneously continue to update and maintain the object-label relationships from past training, which may in turn have "washed out" some of the learning benefits that participants might have accrued from targeting the ambiguous items during the Sampling Phase. Future research could specifically target what types of sampling behavior leads to better learning outcomes, relative to randomly selected training events.

S2: Experiment 2 Practice Phase

At the beginning of the experiment, participants completed a practice phase in which they encountered the two familiar word object stimuli and two novel object-label associations. We introduced this short practice phase to give children experience with the overall structure of the main experiment under less demanding circumstances, using a smaller set of items and mixing familiar and novel items. First, children were exposed to 4 training practice trials similar in structure to the training trials in the main experiment. On each trial, two referents appeared on the screen on either side of the Teddy character and children heard two labels, one for each object, in random order. On the first trial, children always saw the two familiar items (i.e., the penguin and the dog), followed by a second trial in which children saw two novel object-label associations (i.e., an ambiguous labeling event). On the final two training practice trials, children saw each of the familiar items occur with one of the two novel items (permitting the disambiguation of the novel object-label associations). Next, children saw two sampling practice trials, in which children had the opportunity to select which of the four items they wanted to learn about next, followed by four practice test trials, in which participants' knowledge of the items was tested in a 4-AFC recognition test. The procedure for each of these practice trial types mirrored the procedure for the Sampling phase and the Test phase described in the methods section of Experiment 2.

S3: Experiment 2 Supplementary Figures

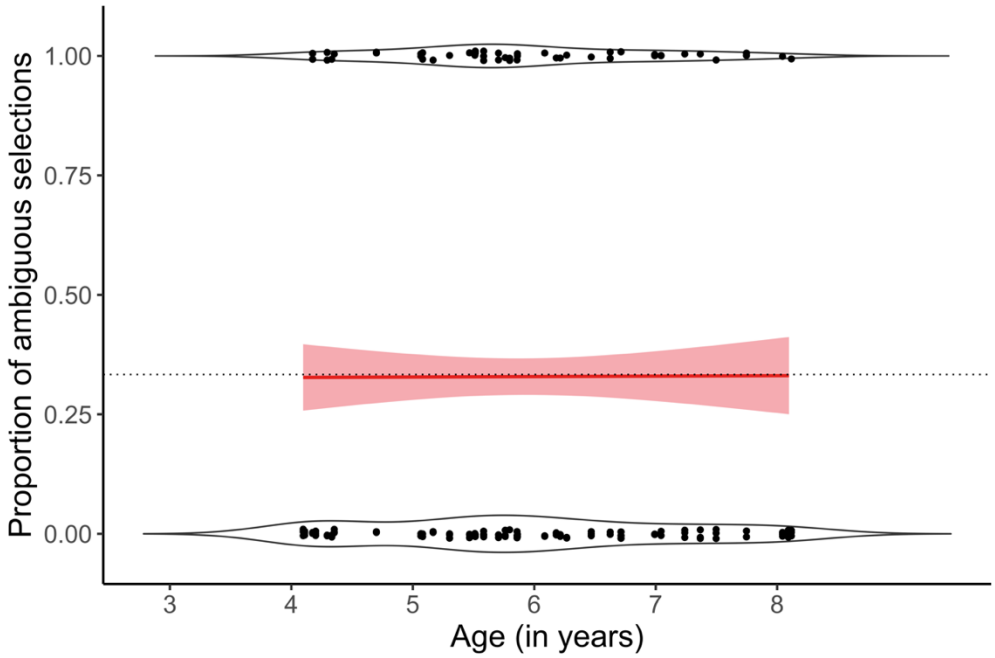


Figure S3.1. Distribution of ambiguous item selections in Experiment 2 across age. Error bands are +1/-1 SEs based on model estimates.

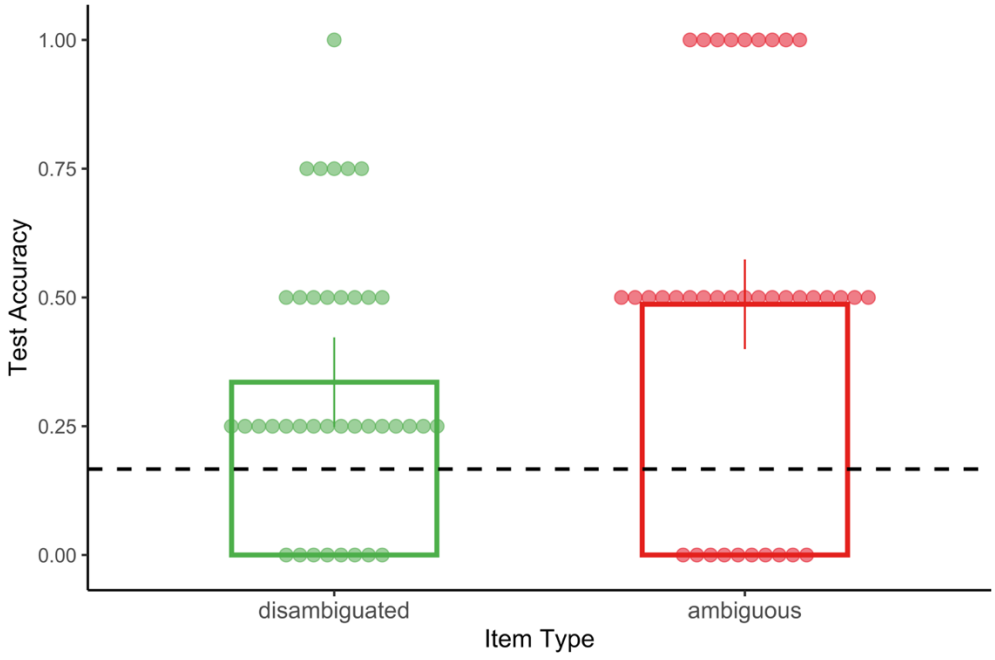


Figure S3.2. Children's test accuracy in Experiment 2, split by Item Type (disambiguated vs. ambiguous). Error bars represent within-participant 95% CIs (Morey, 2008).

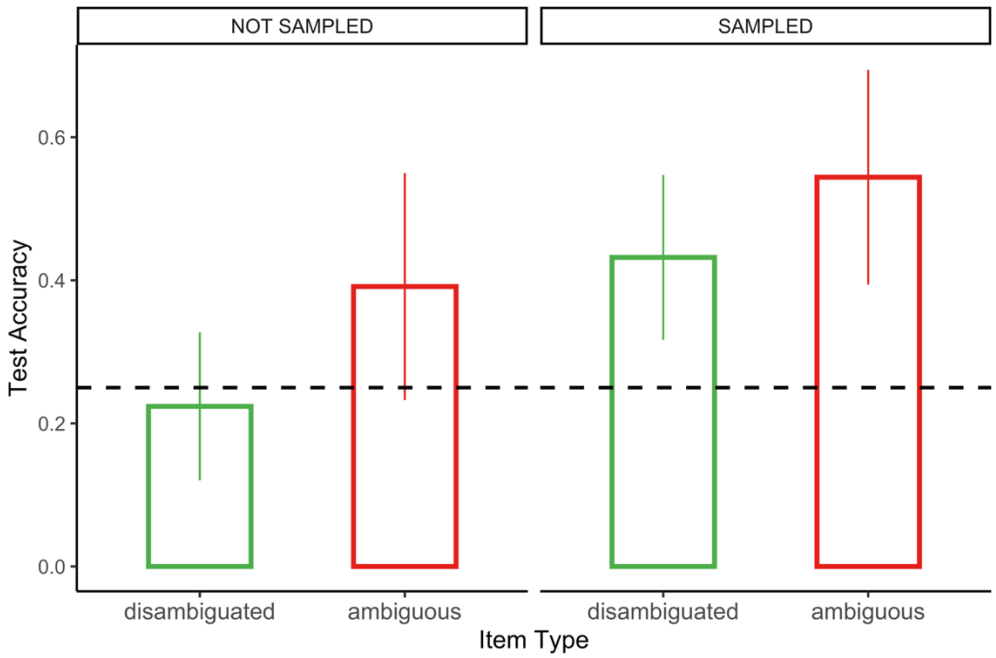


Figure S3.3. Children's test accuracy in Experiment 2, split by Item Type (disambiguated vs. ambiguous) and whether a given test item was selected during the Sampling Phase (not sampled vs. sampled). Error bars represent within-participant 95% CIs (Morey, 2008).

S4: Experiment 3 Supplementary Results

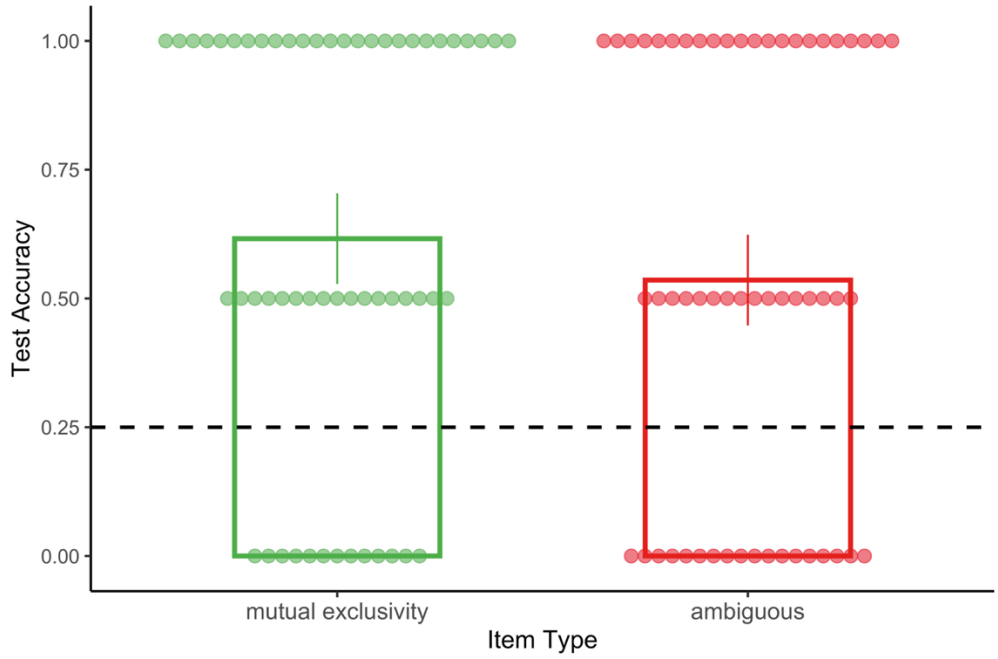


Figure S4.1. Children's test accuracy in Experiment 3, split by Item Type (disambiguated/ mutual exclusivity vs. ambiguous). Error bars represent within-participant 95% CIs (Morey, 2008).

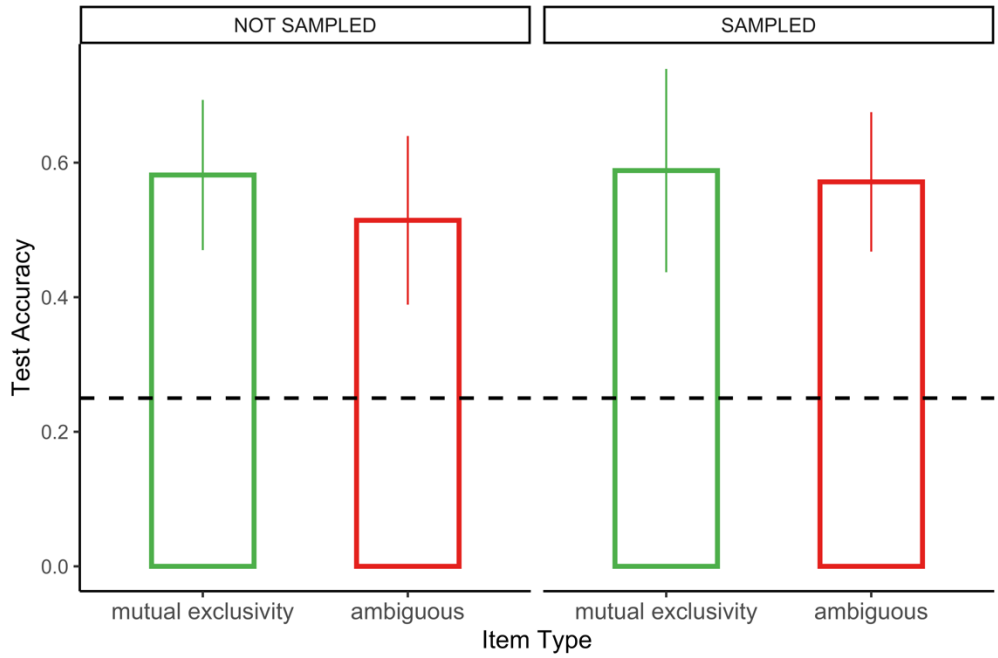


Figure S4.2. Children's test accuracy in Experiment 3, split by Item Type (mutual exclusivity vs. ambiguous) and whether a given test item was selected during the Sampling Phase (not sampled vs. sampled). Error bars represent within-participant 95% CIs (Morey, 2008).